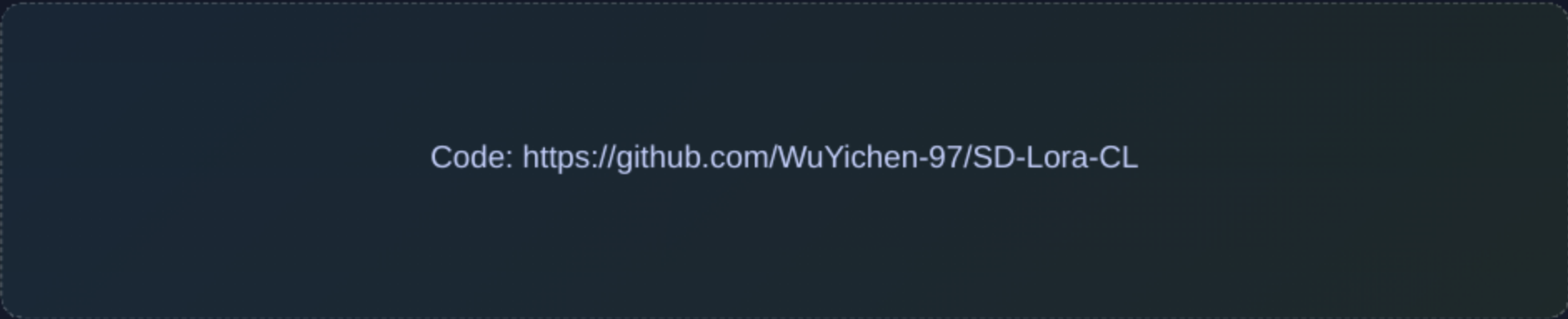


SD-LoRA: Scalable Decoupled LoRA for CL

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- Rehearsal-free
- Inference-efficient
- End-to-end optimized



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Introduction

- Continual Learning: sequential non-i.i.d. tasks; key challenge: catastrophic forgetting.
- Foundation models + prompts/LoRA mitigate forgetting but often need prompt/LoRA selection or rehearsal.
- Scalability goals: rehearsal-free, inference efficiency, and end-to-end optimization.

Our Idea: SD-LoRA

- Incrementally decouple magnitude and direction; keep past directions fixed, learn magnitudes + new direction per task.
- Directly use final composed model at test time (no selection).

Abstract

- CL with foundation models is promising but current prompt/LoRA methods scale poorly (growing pools or rehearsal).
- We propose SD-LoRA: decouples magnitude and direction of LoRA components and learns incrementally without rehearsal.
- Empirical + theoretical analysis: follows low-loss trajectories; converges to overlapping low-loss region → strong stability–plasticity.
- Two parameter-efficient variants: SD-LoRA-RR (rank reduction) and SD-LoRA-KD (knowledge distillation).
- Efficient inference using the final model without component selection; end-to-end trainable; state-of-the-art across benchmarks.

Desirable Properties vs. Prior Work

Method	Rehearsal-free	Inference efficiency	End-to-end optimization
L2P	✓	✗	✗
DualPrompt	✓	✗	✗
CODA-Prompt	✓	✗	✓
HiDe-Prompt	✗	✗	✓
InfLoRA	✗	✓	✓
SD-LoRA (Ours)	✓	✓	✓

SD-LoRA satisfies all three properties without task-ID.

Related Work (Brief)

Continual Learning

- Rehearsal-based, regularization-based, architecture-based.
- Class-incremental setting without task identity at test time.

Foundation Models in CL

- Prompt-tuning with ViTs.
- Some methods require memory or selection.

PEFT

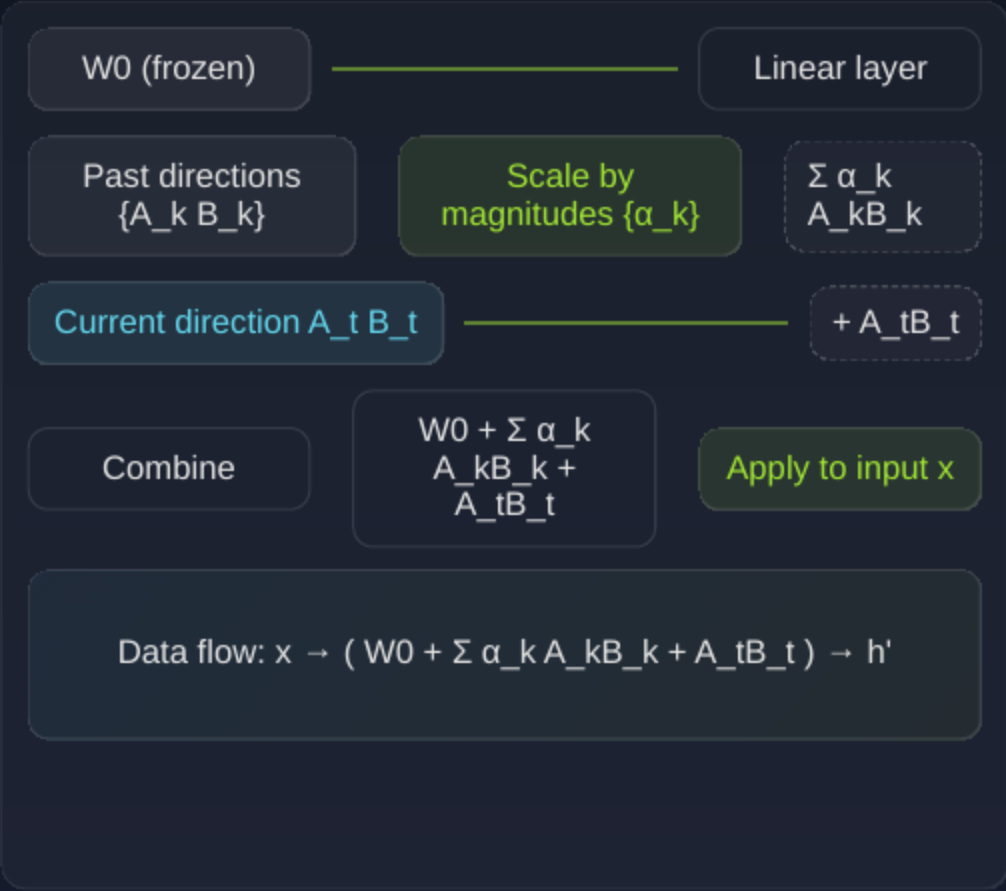
- Adapters, prompts/prefixes, LoRA.
- SD-LoRA: LoRA-based, rehearsal-free for CL.

Preliminaries

- Sequential tasks: $T_1 \dots T_N$; no access to prior data when training T_t .
- Objective: maintain performance on current and past tasks.
- LoRA: $\Delta W = A B$ (low-rank); W_0 frozen; update layer output via $W_0 + AB$.

Figure placeholder: LoRA update vs. SD-LoRA update (Fig. 1)

SD-LoRA: Decoupled Magnitude & Direction



Layer output (visual)

$$h' = (W_0 + \sum \alpha_k (A_k B_k) + (A_t B_t)) x$$

Optimize $\{\alpha_k\}$ and $A_t B_t$; keep W_0 and past directions fixed.

Empirical Findings (1)

Finding 1

- Task-specific fine-tuned weights cluster closer to each other than to W_0 .
- Fix first learned direction; adapt only magnitude $\rightarrow$ outperforms vanilla LoRA in CL.

Fig. 2(a) placeholder: relative distances W^* vs W_0

Fig. 2(b,c) placeholder: accuracy vs. tasks (fixed direction vs. vanilla)

Empirical Findings (2)

Finding 2

- Early directions are heavily reused; new directions gradually diverge.
- Magnitudes α for early tasks increase; later tasks show declining α .

Fig. 3(a) placeholder: LS residual over time

Fig. 3(b,c) placeholder: α evolution across tasks

Empirical Findings (3)

Finding 3

- SD-LoRA traces a low-loss path to a shared low-loss region across tasks.
- Interpolation: SD-LoRA path improves T2 without degrading T1; vanilla LoRA forgets.

Fig. 4(a) placeholder: trajectory illustration

Fig. 4(b,c) placeholder: accuracy along paths

Theoretical Insight

- Gradient descent with small initialization on AB sequentially recovers principal components of an optimal update.
- Explains importance of early directions and decreasing magnitudes.

Theorem placeholder: convergence to $\Delta W[:1]$, $\Delta W[:2]$, ..., $\Delta W[:j]$

Variants: Parameter Efficiency

SD-LoRA-RR (Rank Reduction)

- Stepwise reduce rank r_t for later tasks ($t = \mu, v$).
- Lowers compute/memory with minor accuracy drop.

SD-LoRA-KD (Knowledge Distill.)

- Least-squares fit new direction by span of past ones.
- If residual $\leq \tau$, absorb via magnitude updates; avoid adding directions.

Algorithm 1 placeholder: training loop + RR + KD decision

Experiments: Setup

- Benchmarks: ImageNet-R (N=5/10/20), ImageNet-A (N=10), DomainNet (N=5).
- Backbones: ViT-B/16 pretrained on ImageNet-21K and fine-tuned on ImageNet-1K; also a self-supervised ViT-B/16.
- LoRA in attention (Q,V); r1=10; shared magnitudes across projections.
- Training: Adam, lr=0.008; batch=128; 30 epochs (ImageNet-R), 10 (DomainNet), 20 (others).
- Metrics: Acc and AAA; mean±SE over 5 runs.

Results: ImageNet-R

- SD-LoRA tops Acc/AAA across N=5/10/20.
- Large margins over InfLoRA at N=20 (e.g., +7.68% Acc).
- RR and KD variants closely match SD-LoRA.

Table 2 placeholder: ImageNet-R Acc and AAA across task lengths

Results: ImageNet-A & DomainNet

ImageNet-A (N=10)

- SD-LoRA outperforms prior methods (e.g., >+31% Acc vs HiDe-Prompt).
- AAA also highest.

DomainNet (N=5)

- Consistent SOTA Acc/AAA despite domain diversity.

Table 3 placeholder: Acc and AAA comparisons

Training Dynamics

- Average accuracy improves steadily across tasks on ImageNet-A/R and with the self-supervised backbone.

Fig. 5 placeholder: AAA/Acc vs tasks (multiple datasets/backbones)

Ablation Studies

- Fixed first direction + magnitude tuning already strong → early direction is key.
- Decoupling with single component < SD-LoRA → multiple components + rescaling are critical.
- Removing magnitude rescaling degrades performance → needed for low-loss navigation.

Table 4 placeholder: ablation comparisons

Limitations

- Domain shift brittleness: performance may drop on out-of-distribution domains beyond evaluated splits.
- Sensitivity to early direction quality: later tasks rely heavily on initial directions; poor early learning can propagate.
- Highly dissimilar tasks: shared low-loss region may be weak, reducing gains versus rehearsal-based methods.
- Parameter growth (base SD-LoRA): adds a new direction per task; mitigated by RR/KD but not eliminated in base variant.
- Hyperparameter dependence: RR/KD require thresholds/rank schedules; suboptimal settings may hurt accuracy.

Evidence: Findings 2–3 and Tables 2–3

Efficiency Analysis

- Inference GFLOPs: SD-LoRA matches InfLoRA and avoids selection overhead.
- Trainable params: SD-LoRA-RR reduces LoRA params further.
- No stored samples/features (rehearsal-free).

Table 5 placeholder: GFLOPs, parameters, storage

Open Questions

- How robust is SD-LoRA to stronger domain shifts or distribution drifts between tasks?
- Can we automatically assess and improve early direction quality during CL?
- What principled rank-reduction or KD criteria best balance accuracy and efficiency?
- How does SD-LoRA extend to non-ViT backbones and multimodal foundation models?
- Can we combine SD-LoRA with other PEFT for further gains?
- What theoretical conditions guarantee a shared low-loss region across highly dissimilar tasks?